Recovering Markov dynamics from noised sequences: tree-exact inference and transition learning

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Every number reported here is generated from frozen experimental outputs by the scripts named in Appendix K, which also records the provenance gate each run must pass to be cited. The derivations, the code and the full experimental record are in the accompanying compendium.

Abstract

Over an unrestricted function class, the global minimiser of the denoising objective that trains a diffusion model is not the score of the data distribution: it is the score of the *empirical* distribution, a sum of point masses at the training samples smoothed by the forward noise. A model attaining that target exactly, integrated to zero noise, would reproduce its training set and nothing else, yet diffusion models generate new data. Deciding what a trained network has actually learned requires comparing it against the score it should have learned — and for essentially every distribution of interest that target is unavailable, so a network's error cannot be separated from the difficulty of its target. Tractable analytical models exist, but the most common are Gaussian, whose linear scores cannot probe how an approximation handles genuinely nonlinear posterior structure.

We study a family where the target is computable and non-trivial: a first-order Markov chain on $\mathbb{R}^L$ with continuous, non-Gaussian transitions, variance-matched so that innovation families differ only beyond second moments. Coordinatewise Ornstein–Uhlenbeck noising multiplies the prior by *unary* factors, adding no edges among the latent variables, so the posterior factor graph is still the prior's chain — a tree, on which sum-product is exact. The score is therefore available as an exact functional recursion for a correlated, non-Gaussian model, and numerically to a validated tolerance once the continuous messages are discretised.

We then drop the assumption that the chain's parameters are known and learn them from noised sequences alone. Fisher's identity supplies the exact gradient of the marginal log-likelihood from one complete forward–backward pass, with no differentiation through the recursion, so both gradient ascent and expectation–maximisation are available; for Gaussian innovations the maximisation step is closed form and the two routes reach the same optimum. We prove that single-Gaussian message closure computes *exactly* the LMMSE estimator of the covariance-matched Gaussian model, so the gap between it and full sum-product is precisely the price of discarding everything beyond second moments. Estimating a transition *density* rather than its parameters exposes a convergence-rate asymmetry between the correlation and shape coordinates, to which two mechanisms contribute: the innovation mixture's own latent label, which makes the shape slow even on clean transitions, and the observation channel, which compounds it and removes far more Fisher information about shape than about correlation.

Against an *unstructured* network trained by denoising score matching on identical data, with the optimisation budget of both arms selected on a held-out bundle, the structured estimator attains between seven and eighteen times lower pointwise denoising error, at equal training-set size, across 8 sizes — a ratio of errors at fixed data, not a reduction in the data required to reach a fixed error. A large fraction of that margin disappears once the baseline is given locality and weight sharing: against a window head, selected by screening over three candidate architectures and measured at the same headline protocol, all-site scored, the advantage falls to 2.3–$6.1\times$, increasing across the sizes tested rather than falling. The comparison is deliberately asymmetric in prior information — the estimator is handed the Markov factorisation, the autoregressive form and a low-dimensional innovation family, the networks none of them — so it measures the worth of that structure on this family, not a general ranking of the two approaches.

1 Introduction

Generative diffusion produces data $\mathbf{a} \in \mathbb{R}^L$ from a distribution P_0 known only through samples, by noising and then reversing [1–3]. Reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$, which is learned by regression against the posterior mean of the clean signal.

The puzzle that motivates this. A network never sees P_0. It sees M samples, the empirical measure $\hat{P}_0 = M^{-1} \sum_\mu \delta(\cdot - \mathbf{a}^\mu)$, and over an unrestricted function class the global minimiser of the denoising objective is the score of *that*. A model attaining it exactly, integrated to zero noise, would have a reverse process whose fixed points are the training samples and would generate nothing else. Diffusion models nonetheless generate new data, and when they memorise rather than generalise is an active question [4, 5].

Measuring it is the difficulty: deciding what a network has learned means comparing it against the score it *should* have learned, and for realistic data that object is unavailable, so a large error may indicate a poor model or a hard target with nothing to distinguish them. The tractable alternatives are poor probes of *structure*. A Gaussian score is linear, so any question about exploiting correlation has a degenerate answer; a mixture score is nonlinear through its responsibilities — in the mixture index, not in any dependence between coordinates.

Why a Markov chain. Two reasons. Diffusion is applied to sequential data — video, audio, time series — so a model of correlation along a sequence is the relevant caricature. And studying how structure is exploited requires data that has structure, of a kind one can dial. A first-order Markov chain with continuous non-Gaussian transitions supplies both, and the variance-matching of §3 makes the innovation family a controlled knob: Gaussian, Laplace, Student, uniform and bimodal chains share a covariance exactly and differ only beyond it.

Why it is tractable. Coordinatewise noising contributes one *unary* factor per site. Unary factors reweight node potentials without creating edges among the latent variables, so conditioning leaves the factor graph a chain — a tree, on which sum-product returns exact marginals. The exact score is therefore computable for a correlated, non-Gaussian model, and the only obstruction is that the messages are functions on $\mathbb{R}$ and must be represented. The Gaussian case, where the answer is known in closed form, is what validates that representation before it is used where the answer is not known.

The question this makes askable. If the data is Markov, is it *easier to learn*? Sum-product reaches the exact score through very simple operations, and it is Markovianity that permits this. So we drop the assumption that the chain's parameters are known: keep the recursion, treat the kernel as unknown, start from a random initialisation, and ascend the likelihood on a finite sample, with Fisher's identity supplying the exact gradient. The resulting estimator is compared against networks trained by denoising score matching on identical data.

Rather than survey what the model can do, we ask one question — *given only sequences seen through the noising channel, what can be recovered?* — at three levels of difficulty, deriving the answer at each before measuring it:

	what is unknown	what is derived	§
0	nothing — the kernel is given	the posterior mean, in closed form	4
1	two numbers (ρ, q)	exact gradient; closed-form maximisation	5
2	a transition *density*	a second latent; a rate asymmetry	6

The three share one model, one channel and one estimator, differing only in how much of the transition is withheld; each adds exactly one difficulty to the one below it.

Contributions.

- A denoising reference for a continuous, non-Gaussian, correlated model, with truncation and quadrature error separated and measured, validated against a closed form and against brute-force enumeration (§4).
- An identification of what single-Gaussian message closure computes here: exactly the LMMSE estimator of the covariance-matched Gaussian model (Proposition 1), proved directly rather than argued asymptotically.
- Estimation of the kernel from noised sequences alone, with the exact gradient from one complete forward–backward pass and no differentiation through the recursion (§5–§6).
- A measurement of what supplying that structure is worth against trained denoisers given none of it, both arms' budget selected on held-out data (§7).

2 Related work

Diffusion and score matching. The construction is due to Sohl-Dickstein et al. [1], developed by Ho et al. [2] and given its continuous-time form by Song et al. [3]; training targets the score by denoising score matching [6, 7], and that the regression target is a posterior mean is Tweedie's formula [8–10].

Statistical mechanics, and the memorisation question. The high-dimensional behaviour of these models is analysed as a phase-transition problem [4, 5, 11]. The puzzle above is attacked within it through training dynamics, where the time to memorise grows with sample size while the time to generalise does not [12, 13]; through the learned representation, geometry-adaptive rather than sample-specific [14]; through the spectrum of the empirical score [15]; and in solvable settings [16–18], with Garnier-Brun et al. [19] exhibiting a phase where test loss still falls while samples drift anomalously close to the training set. Every one measures a network against a target unavailable in closed form — which is what the model below supplies.

Tractable structured data, and architectures that implement inference. Hierarchical generative models supply tractable non-Gaussian families with a phase structure [20], and the same construction can be run backwards to read latent structure off real data [21] or to date the emergence of compositional generalisation during training [22]. Two results bear directly on what we leave open: Mei [23] reads U-Nets on such models as implementing belief propagation, and Garnier-Brun et al. [24] shows transformers learning the corresponding inference on hierarchically filtered data. Both concern *tree*-structured data, where the natural depth is the tree's. A chain is a degenerate tree — depth equal to its length, width two — so the architectural implication differs, which is what §7 sets up rather than settles.

Graphical models and inference. Sum-product and its exactness on trees are standard [25–28]; on a chain the recursions are forward–backward [29, 30], reducing to Kalman and RTS in the Gaussian case [31, 32]. The estimation machinery is EM [33], its conditional-maximisation variants [34, 35], and Fisher's identity [36–38].

What is not new here. The numerics are classical too: messages on a grid, integrated by quadrature, is Kitagawa's non-Gaussian state-space smoother [39], alongside Gaussian-sum [40], expectation-propagation [41] and Monte Carlo [42] alternatives [see 43]. Grid BP is Kitagawa's method at each diffusion time; Rung 1 is maximum likelihood for a discretised continuous-state HMM, whose identifiability under noisy observation is itself studied [44]. The contribution is not the estimator but what this model permits: a score known across the whole schedule at a controlled tolerance, turning "what does an approximation cost" into a measurement.

3 Setup

Notation. L is the chain length, which is also the ambient dimension, and M is the number of observed sequences; the diffusion literature's n and d translate as $n \to M$, $d \to L$. Vectors are bold lowercase and matrices bold uppercase, so $\mathbf{a} \in \mathbb{R}^L$ is a whole clean chain and a_i its i-th site. K is always the transition kernel and N_g always the number of quadrature points (Appendix A).

Forward process. With η standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \;=\; -\mathbf{x} + \sqrt{2}\,\eta(t), \qquad \mathbf{x}(t) \;=\; e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})\,. \tag{1}$$

The reverse-time convention is fixed once in Appendix A and used without variation.

Prior. P_0 factorises as

$$P_0(\mathbf{a}) \;=\; \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \tag{2}$$

with φ an innovation density of mean zero and variance q, and $|\rho| < 1$. We draw $a_1 \sim \mathcal{N}(0,1)$ and set $q = 1 - \rho^2$, which makes $\mathrm{Var}(a_i) = 1$ at every site and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly, *whatever the shape of* φ. Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share the same covariance and differ only beyond second moments. The chain is covariance-stationary but not strictly stationary; Appendix E states the difference, and shows why it collapses in the Gaussian case.

The score is a posterior mean. From (1), P_t is a Gaussian convolution of P_0. Differentiating under the integral and dividing by $P_t(\mathbf{x})$ — the Gaussian factor over $P_t(\mathbf{x})$ is exactly the posterior density $P(\mathbf{a} \mid \mathbf{x}, t)$ — gives

$$S(\mathbf{x}, t) \;=\; -\,\frac{\mathbf{x} - e^{-t}\,\langle \mathbf{a} \rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a} \rangle_{x,t} = \int \mathrm{d}\mathbf{a}\; \mathbf{a}\, P(\mathbf{a} \mid \mathbf{x}, t). \tag{3}$$

Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [8–10] and underlies denoising score matching [6, 7]. For squared loss $\langle \mathbf{a} \rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every estimator below is scored against.

Conditioning preserves the chain. Conditioning (2) on $\mathbf{x}$ gives

$$P(\mathbf{a} \mid \mathbf{x}, t) \;\propto\; \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}) \prod_{i=1}^{L} \ell_t(x_i \mid a_i), \qquad \ell_t(x_i \mid a_i) \propto \exp\!\left[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right]. \tag{4}$$

The likelihood factors are *unary*: the forward noise is independent across coordinates, so each observation touches one latent variable. They reweight node potentials and add no edges, so the posterior factor graph is the prior's chain (drawn in Appendix D).

A *tree* is a connected acyclic graph — there is exactly one path between any two nodes. A chain is the simplest tree, the one whose nodes have degree at most two. On a tree, sum-product returns the exact marginals [25–27]: with

$$\begin{aligned}
\overrightarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i-1}\; K(a_i \mid a_{i-1})\, \ell_t(x_{i-1} \mid a_{i-1})\, \overrightarrow{m}_{i-1}(a_{i-1}), \\
\overleftarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i+1}\; K(a_{i+1} \mid a_i)\, \ell_t(x_{i+1} \mid a_{i+1})\, \overleftarrow{m}_{i+1}(a_{i+1}),
\end{aligned} \tag{5}$$

initialised by $\overrightarrow{m}_1 = \mu$ and $\overleftarrow{m}_L \equiv 1$, the single-site marginal is $P(a_i \mid \mathbf{x}, t) \propto \overrightarrow{m}_i(a_i)\ell_t(x_i \mid a_i)\overleftarrow{m}_i(a_i)$ and the pairwise marginal on edge $(i-1, i)$ is $\propto \overrightarrow{m}_{i-1}\ell_t(x_{i-1} \mid a_{i-1})K(a_i \mid a_{i-1})\ell_t(x_i \mid a_i)\overleftarrow{m}_i$. The local likelihood is absorbed *before* propagation, so $\ell_t(x_i \mid a_i)$ appears exactly once in each marginal. These are the forward–backward recursions of hidden Markov models [29, 30] written for a continuous state; when everything is Gaussian they reduce to the Kalman filter and RTS smoother [31, 32].

Two cautions, both expanded in Appendix C. Sum-product is exact on the tree, but a message is a *function* on $\mathbb{R}$ — forced by the state space, not chosen — so an implementation must represent one, and the truncated grid we use makes the recursion exact only for a finite-state approximation whose error §4 measures. And Monte Carlo enters this note in exactly one place, when a statistic is computed from generated samples; every other number comes from deterministic quadrature and carries discretisation error but no sampling error.

Cost. The discretised posterior has N_{g}^L joint configurations, so one site marginal by direct summation fixes that site and sums over the other $L-1$: $O(N_{\mathrm{g}}^{L-1})$ per marginal, about 10^{80} terms at $N_{\mathrm{g}} = 401$, $L = 32$. The recursion replaces it with $2(L-1)$ matrix–vector products: $O(LN_{\mathrm{g}}^2)$ time per sequence and $O(LN_{\mathrm{g}})$ storage for the messages, plus one shared $O(N_{\mathrm{g}}^2)$ transition matrix. Exponential in the chain length becomes linear in it and quadratic in the representation size (Appendix D).

4 Rung 0 — the kernel is known

With K given, (5) returns the posterior mean and hence, through (3), the score. Two things must be established before anything is built on it: that the discretisation is controlled, and what the standard Gaussian approximation actually computes.

The discretisation, controlled on two axes. Two error sources are separated by construction: *truncation*, mass outside $[-A, A]$ discarded rather than transported, controlled by A and essentially independent of N_{g}; and *quadrature*, second order in h, and no better than second order for the Laplace family, whose innovation has a discontinuous derivative at the origin. Each is measured with a diagnostic that isolates it, since a resolution sweep at fixed A cannot rule out truncation and a domain sweep at fixed N_{g} cannot rule out quadrature (Figure 3, Appendix G). The working configuration $N_{\mathrm{g}} = 401$, $A = 8$ is read off those curves, and puts both diagnostics far below every effect reported here.

Validation where the answer is known. Take $\varphi = \mathcal{N}(0, q)$. Then $\mathbf{a}$ is jointly Gaussian with $(\mathbf{\Sigma}_0)_{ij} = \rho^{|i-j|}$, and so is $\mathbf{x}$, with $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$, giving

$$\langle \mathbf{a} \rangle_{x,t} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}, \qquad S(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}. \tag{6}$$

The clean precision $\mathbf{\Sigma}_0^{-1}$ is tridiagonal, and the posterior precision $(e^{-2t}/\Delta_t)\mathbf{I} + \mathbf{\Sigma}_0^{-1}$ stays tridiagonal at every t: being Markov is a statement about the precision, not the covariance, which is dense. At $N_{\mathrm{g}} = 401$, $A = 8$ the discretised recursion reproduces (6) to 9.2×10^{-15} relative, and agrees with brute-force enumeration of the discretised posterior — summing over all N_{g}^L configurations on a short chain, sharing no code with the message passing — to 1.6×10^{-14} on posterior means and 1.8×10^{-15} on the log-evidence. Neither check involves non-Gaussian machinery: the pipeline is verified where the answer is known before being used where it is not.

What the standard Gaussian approximation computes. A common approximation propagates only the first two moments of each message. Under $a_i = \rho a_{i-1} + \varepsilon_i$, a message with mean m and variance v transforms as

$$\text{forward:} \quad m \mapsto \rho m, \;\; v \mapsto \rho^2 v + q; \qquad \text{backward:} \quad m \mapsto m/\rho, \;\; v \mapsto (v+q)/\rho^2, \tag{7}$$

the backward map being the forward one inverted because that recursion integrates over the output variable (it requires $\rho \neq 0$). The propagation uses only the mean and variance of φ.

Proposition 1 (Gaussian closure computes the LMMSE estimator)**.** *Let $\Pi_{\mathrm{G}} f$ be the Gaussian density matching the mean and variance of $f/\int f$, and let* moment-projected sum-product *be* *the forward–backward recursion applying Π_{G} to each message immediately after every transition integral, multiplying by the site's unary likelihood before the next. Take the chain* ([2]) *with $a_i = \rho a_{i-1} + \varepsilon_i$, $\rho \neq 0$, Gaussian unary likelihoods, $\mathbb{E}[a_1] = 0$, an innovation law φ with mean zero and finite variance q, and an initial message Gaussian or projected by Π_{G}. Then it returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\ \mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\mathbf{\Sigma}_0 \left(e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}\right)^{-1}\mathbf{x}, \tag{8}$$

and returns the same estimator for every innovation law sharing the moments $(0, q)$.

We call (8) the *Gaussian second-order baseline*. On this model, message closure, moment projection and outright replacement of the prior by its covariance-matched Gaussian all coincide; they separate for richer message families. We do not call the method AMP: a degree-two chain supplies no large-degree central-limit justification for that terminology, and Proposition 1 is proved directly rather than argued asymptotically, by induction over the two recursions (Appendix F, with Remark 2 on why each hypothesis is needed and on the fact that the grid plays no part in the statement). The zero-mean hypothesis is load-bearing in practice as well as in the proof: at $\mathbb{E}[a] = 1.3$ the estimator (8) departs from the true LMMSE by 0.65 in maximum absolute value.

Proposition 1 is what makes the gap between (8) and grid BP interpretable: it is not an artefact of a message representation, it is exactly the price of replacing the prior by its covariance twin. That gap locates where non-Gaussianity matters (Figure 1), and it shrinks as t grows — because the noised marginal P_t Gaussianises and Tweedie's prefactor α_t/Δ_t attenuates the score's sensitivity to structure beyond second moments, *not* because the latent posterior becomes Gaussian. It does not: as $t \to \infty$ the observation becomes uninformative and $P(\mathbf{a} \mid \mathbf{x}, t)$ approaches the prior, the non-Gaussian object at issue.

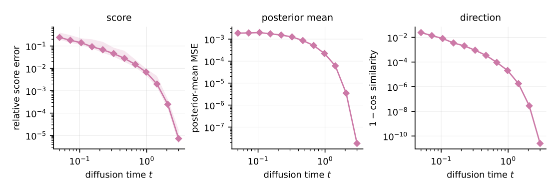

Figure 1: The cost of the Gaussian second-order baseline, against grid BP under the true kernel. Laplace chain, $\rho = 0.85$, $L = 32$, median over seeds with the 10–90 percentile band. By Proposition 1 this gap is exactly the price of replacing the prior by its covariance-matched Gaussian. It vanishes as t grows, because the noised marginal Gaussianises and Tweedie's prefactor α_t/Δ_t attenuates the score's sensitivity to structure beyond second moments — not because the latent posterior itself becomes Gaussian, which it does not.

5 Rung 1 — recovering two numbers

Now K is unknown. Take the simplest case first: Gaussian innovations, so the kernel is $K_\theta(a' \mid a) = \mathcal{N}(a' - \rho a, q)$ and $\theta = (\rho, q)$ is two numbers. We observe only noised sequences $\mathbf{x}^\mu$, $\mu = 1, \ldots, M$, and write $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(\mathbf{x}^\mu)$ for the marginal log-likelihood.

The gradient needs no backpropagation. The obvious route is gradient ascent on $\mathcal{L}$. That looks expensive — $\mathcal{L}$ is defined through the recursion, so differentiating it appears to require backpropagating through $2(L-1)$ message updates. It does not. Fisher's identity [36–38] gives

Proposition 2 (Exact gradient from one forward–backward pass)**.** *Fix the grid $u_1, \ldots, u_{N_{\mathrm{g}}}$ and let K_θ be the induced finite-state transition matrix, differentiable in θ. With $\mathbf{\Xi}$ the expected transition mass, indexed* ***child-first*** *to match $[K_\theta]_{kl} = K_\theta(u_k \mid u_l)$,*

$$\mathbf{\Xi}_{kl} \;=\; \sum_{\mu=1}^{M}\sum_{i=2}^{L} P(a_i = u_k,\ a_{i-1} = u_l \mid \mathbf{x}^\mu)\,, \tag{9}$$

the gradient of the discretised *marginal log-likelihood is*

$$\nabla_\theta \mathcal{L}(\theta) = \sum_\mu \mathbb{E}_\theta[\nabla_\theta \log P_\theta(\mathbf{a}, \mathbf{x}^\mu) \,|\, \mathbf{x}^\mu] = \big\langle \mathbf{\Xi}(\theta),\ \nabla_\theta \log K_\theta \big\rangle. \tag{10}$$

Both Ξ and the evidence come from a single complete forward–backward pass — one forward and one backward recursion, not one directional sweep — so (10) is of the same asymptotic order as evaluating the likelihood, with no additional message pass and no backpropagation through the recursion. It is not free: accumulating Ξ over all (k, l) dominates the constant at large N_{g}.

Sum-product is not a computation graph to backpropagate through; it supplies the conditional expectation, and one forward–backward pass yields the exact gradient of the discretised marginal likelihood (Remark 3). We verify (10) against finite differences to $\sim 10^{-9}$; that check shares the forward recursion with what it validates, so the independent controls are brute-force posterior enumeration on a tiny grid and the closed-form Gaussian gradient (Appendix K).

Ξ is the continuum analogue of Baum–Welch's expected transition counts [29], and it is sufficient *for the current* $\mathcal{Q}$ only — not a sufficient statistic of the data. Appendix K states the conditions and the cost.

Two routes to the same place. Equation (10) licenses gradient ascent; EM instead increases $\mathcal{Q}$. For the Gaussian kernel the maximisation step is closed form — profiling $\mathcal{Q}$ gives the belief-weighted Yule–Walker equations

$$\rho \;=\; S_{01}/S_{00}, \qquad q \;=\; (S_{11} - \rho\, S_{01})/W, \tag{11}$$

with $S_{ab} = \sum_{kl} \Xi_{kl}\, u_k^a u_l^b$ and $W = \sum_{kl} \Xi_{kl}$ — so this rung has *exact* EM, not a generalised variant. The derivation is Appendix H.

Both routes are implemented and neither dominates. On this smooth kernel gradient ascent reaches EM's optimum to 1.6×10^{-9} nats, but it needs a tuned step size and larger steps diverge; EM needs no step size and its ascent is monotone by construction [33, 34]. That is the whole content of the comparison, and it is why every rung above uses EM. The one place gradient ascent wins is a kernel whose $\mathcal{Q}$-maximiser is a lattice artefact of the discretisation, which is a statement about the grid rather than about the algorithms (Appendix H).

Identifiability, at this rung, is a theorem. (11) inverts the map $(\rho, q) \mapsto (S_{01}/S_{00}, \dots)$ explicitly, and Gaussian convolution at finite t is injective on distributions — the characteristic function acquires a factor $e^{-\Delta_t \|\xi\|^2/2}$, which has no zeros, and $e^{-t} > 0$ — so P_t determines P_0 and P_0 determines (ρ, q). Nothing is assumed here. That changes at Rung 2, and the change is worth marking rather than burying: see §6.

What is measured. Recovery of (ρ, q) from initialisations away from the truth, and the rate at which the error falls with M. Theory predicts $M^{-1/2}$, the ordinary parametric rate, since this is maximum likelihood in a correctly specified two-parameter model. Over $M = 64$ to 1024 at sixteen replicates, the innovation-variance error falls from 2.7×10^{-2} to 6.4×10^{-3}, a fitted log–log slope of -0.52 against the predicted -0.50 ($r = -0.99$, monotone). The ρ error falls monotonically over the same range but more slowly, at -0.27; we report the discrepancy rather than explain it, since at these budgets ρ is already accurate to 5×10^{-3} and we have not established that the residual is sampling error rather than a floor.

6 Rung 2 — recovering a density

Free the innovation shape. Let

$$K_\theta(a' \mid a) \;=\; \sum_{c=1}^{C} \pi_c\, \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)\,, \tag{12}$$

in which **both** the autoregressive coefficient and the innovation density are free. At $C = 8$ this is ρ, seven weights after the simplex constraint, eight locations and eight scales. Two things change at once, and they are worth separating.

A second latent variable. At Rung 1 the complete-data problem — knowing $\mathbf{a}$, fit K — was a two-parameter regression with a closed-form solution. Here, knowing $\mathbf{a}$ is not enough: the complete-data problem is itself a mixture fit, carrying the component label as a second latent variable, so the maximisation step has no closed form. We alternate the (π, ν, s^2) and ρ blocks, each closed form given the other, for up to sixteen inner sweeps or until the per-edge gain in $\mathcal{Q}$ falls below 10^{-8}. The algorithm is therefore **generalised EM** of ECM type [33, 35]: every step increases $\mathcal{Q}$ without maximising it, which preserves monotone ascent [34] but not the exact-maximiser property. We do not call it exact EM.

Misspecification. When the truth is Laplace and the family is the mixture (12), what is recovered is a *density*, not a parameter vector, and the mixture is a flexible representation of it rather than the truth in disguise. Identifiability weakens correspondingly: finite mixtures are identifiable up to label permutation on the interior of the parameter set — all $\pi_c > 0$, the pairs (ν_c, s_c) distinct [45, 46] — which is exactly where an over-specified C need not stay. The condition is monitored (minimum component mass, minimum width in grid spacings) rather than assumed. This sits between Rung 1's theorem and what could be said for a kernel left unconstrained in shape, where even conditional identifiability would not be available.

Why the fourth cumulant, and not some other summary. A density needs a scalar summary to be tracked, and the choice is forced rather than free. Every innovation family used here is symmetric, so all odd cumulants vanish: $\kappa_3 = 0$ identically. The variance is matched by construction, $q = 1-\rho^2$, so κ_2 is fixed across families and carries no discriminating information at all. The first cumulant that *can* differ between a Laplace innovation and its covariance twin is therefore the fourth. Excess kurtosis is not a preference; at matched variance and under symmetry it is the leading coordinate in which non-Gaussianity can appear.

The coordinates do not converge at the same rate. This is the substantive finding of the rung, and it is a statement about the optimiser rather than about information.

Remark 1 (Two different slownesses, kept apart)**.** *On* clean *chains, with no missing information whatsoever, ρ and q are converged after a single maximisation step — with clean data Ξ is the empirical pair distribution and does not depend on θ at all — while the innovation shape still takes tens of iterations. That slowness belongs to the mixture's inner latent label, the extra difficulty introduced above. It is not caused by the channel.*

Through the channel both coordinates slow down further, and that *is the channel's doing: it is the missing-information mechanism of Dempster et al. [33] appearing exactly where the theory puts it, in the convergence rate rather than in the attainable accuracy.*

The two effects are separable and we separate them, because conflating them produces a false conclusion — an apparent information *penalty that is really a rate at a fixed iteration budget. Both arms recover the truth when run to convergence.*

The score is recovered faster than the density that induces it. Reporting recovery through the induced score alone is generous: the channel has already blurred the innovation law by the time the score is evaluated. Measured directly, by Hellinger distance between kernel columns, the density lags. Across $M = 32$ to 2048 the relative score error falls by a factor of 4.5 while the transition distance falls by 2 — a sixty-fourfold increase in data buys much less at the density level than the pointwise number suggests, so any claim that the *law* was recovered should be stated on the law. The figures, and what the scalar summary does and does not mean for a conditional kernel, are in Appendix O.

The practical consequence is that no convergence test can be trusted to select the reported model: ρ settles roughly an order of magnitude sooner than the shape, so a ρ-trace — or a flat likelihood — looks textbook-converged over a kernel nowhere near converged in the coordinate every non-Gaussian claim depends on. Every reported estimator is therefore selected on a disjoint validation bundle, at every rung (Appendix K). The fitted density, the settling iterations and the clean-against-channel comparison are Appendix I.

7 What the structure buys

The three levels above establish what is recoverable. This section asks what the structural assumption is worth, and is deliberately the shortest in the note. As stated in the abstract the comparison is asymmetric by construction, so what follows measures the value of that structure and says nothing about neural denoisers in general. The baseline in Table 1 is a fully connected network trained by denoising score matching in both standard parameterisations, scored against the deterministic grid-BP reference on held-out sequences — carrying neither weight sharing nor a locality prior. Below we replace it with one that carries both.

Table 1: Relative denoising error against the deterministic grid-BP reference under the true kernel, averaged over the noise schedule; 16 seeds $\pm$ one standard error, aggregated per seed. Both arms select their optimisation budget on a disjoint validation bundle. EM–BP is more accurate in 1,518 of 1,520 cells. Protocol, cell exclusions and the budget calibration are in Appendix K.

sequences M	network	EM–BP (24 free)	ratio
32	0.4031 ± 0.0059	$\mathbf{0.0589 \pm 0.0026}$	7.0
64	0.3285 ± 0.0026	$\mathbf{0.0501 \pm 0.0041}$	7.8
128	0.2759 ± 0.0024	$\mathbf{0.0343 \pm 0.0017}$	8.8
256	0.2268 ± 0.0010	$\mathbf{0.0258 \pm 0.0014}$	9.7
512	0.1878 ± 0.0006	$\mathbf{0.0219 \pm 0.0011}$	9.0
1024	0.1569 ± 0.0006	$\mathbf{0.0162 \pm 0.0012}$	12.5
2048	0.1326 ± 0.0005	$\mathbf{0.0132 \pm 0.0008}$	15.5
4096	0.1171 ± 0.0004	$\mathbf{0.0094 \pm 0.0007}$	17.6

Table 1 gives a ratio between 7.0 and 17.6, larger at the largest sample sizes but *not* monotone — it falls from 9.7 at $M = 256$ to 9.0 at $M = 512$ before rising — so we claim no trend. It is also the largest of several defensible aggregations of these cells (Appendix P). Each arm often selects its largest offered budget at the larger sizes, which reports only that validation error is still falling: tripling both budgets at $M = 2048$ moves the ratio by -0.16 ± 0.18. We observe no crossing between the two curves and therefore quote no data-equivalence factor.

How much of this is the architecture? A ratio against a fully connected network cannot separate knowing the Markov structure from sharing weights across sites, and the second is the cheaper explanation. Measured at the headline protocol — every site scored, both arms selected on a disjoint validation bundle, deployed and provenance-gated (Appendix N) — against a weight-shared window head the advantage is 2.3–6.1$\times$, increasing rather than falling with M: EM–BP is more accurate in every one of the 192 scored cells. The window head was chosen by screening against a dilated convolutional stack and a bidirectional message-passing network, both able in principle to propagate across the whole chain, and beat neither: on this family, at these sizes, the strongest amortised-inference baseline tested is the simplest one screened. So the scoped statement is: *on this family, at these sizes, and with each arm's budget chosen on held-out data, the structured estimator attains lower pointwise denoising error at equal training-set size than the networks we trained — by roughly an order of magnitude against a generic function approximator, and by* 2.3–6.1$\times$ *against a baseline that shares weights across sites.*

8 Limitations and outlook

The model assumes exact first-order Markov structure, and under a non-Markov prior the estimator is misspecified in a way more data does not repair: a global latent adds a rank-one term a chain absorbs, while long-range precision coupling is not low-rank. Nine further scope conditions are in Appendix L, and the outstanding measurement is Appendix N. The open question is architectural: which architectures exploit Markovianity, and at what depth [23, 24].

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A Notation and diffusion conventions

A.1 Symbols

symbol	meaning	symbol	meaning
L	chain length = ambient dimension	$K,\ K_\theta$	transition kernel
M	number of observed sequences	φ	innovation density
N_{g}	number of quadrature grid points	ρ	autoregressive coefficient
$u_1, \ldots, u_{N_{\mathrm{g}}}$	grid points	q	innovation variance
h	grid spacing	C	mixture components
A	grid half-width, domain $[-A, A]$	μ	initial law $\mu(a_1)$
t	diffusion time	ℓ_t	unary likelihood factor
$\Delta_t = 1 - e^{-2t}$	noise variance	$\overrightarrow{m},\ \overleftarrow{m}$	forward, backward message
$P_0,\ P_t$	clean, noised distribution	θ	kernel parameters
$S(x,t)$	score $\nabla_x \log P_t(x)$	$\mathcal{Q}$	EM auxiliary function
$\langle a \rangle_{x,t}$	posterior mean of a given x	$\boldsymbol{\Xi}$	expected transition statistic
η	standardised Gaussian white noise	$\mathcal{L}$	marginal log-likelihood
$t_{\min},\ t_{\max}$	reverse-integration endpoints	$\widehat{\mathbf{a}}_{\mathrm{LMMSE}}$	LMMSE estimator

Two symbols are worth pinning down, because neighbouring literatures use them differently. K **always means the transition kernel** and never a grid size; the grid size is always N_{g}. The ambient dimension is written L because it *is* the chain length and every graphical-model statement is indexed by sites; the diffusion literature more often writes n for the sample count, which here is M.

A.2 Time orientation and the reverse SDE

Convention, used without variation. η is standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, so every $\sqrt{2}$ is written explicitly.
Forward, t running from 0 to $t_{\max}$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -x + \sqrt{2}\,\eta(t), \qquad x(t) = e^{-t}a + \sqrt{\Delta_t}\,z, \qquad \Delta_t = 1 - e^{-2t}, \quad z \sim \mathcal{N}(0, I)\,.$$

Reverse, in the elapsed-time variable $\tau = t_{\max} - t$ running from 0 to $t_{\max} - t_{\min}$, so that increasing τ means decreasing t:

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = x + 2\,S(x, t_{\max} - \tau) + \sqrt{2}\,\eta(\tau).$$

Probability-flow ODE, same marginals, deterministic: $\mathrm{d}x/\mathrm{d}\tau = x + S(x, t_{\max} - \tau)$.
Initialisation $x(\tau{=}0) \sim \mathcal{N}(0, I)$; termination at $\tau = t_{\max} - t_{\min}$.

The reverse drift follows from the time-reversal theorem for diffusions [47, 48]: for a forward SDE $\mathrm{d}x = f\,\mathrm{d}t + g\,\mathrm{d}W$ the time-reversed process has drift $f - g^2 \nabla \log p_t$ and the same diffusion coefficient. With $f = -x$ and $g = \sqrt{2}$ the backward-in-t drift is $-x - 2S$, and rewriting in τ flips the sign to give the box above. Because unit-variance data keep $\mathrm{Var}(x_t) = 1$, initialising at $\mathcal{N}(0, I)$ incurs a bias $e^{-2t_{\max}}$, which is 2.5×10^{-3} at $t_{\max} = 3$. The implementation is `src/reverse.py`, whose module docstring states the same convention; the sampler uses a geometric time grid, denser at small t, because the reverse drift stiffens like $1/\Delta_t$ there.

B The score–posterior identity

Differentiation under the integral. $P_t(x) = \int \mathrm{d}a\, P_0(a)\, g_{\Delta_t}(x - e^{-t}a)$ with $g_{\Delta_t}(u) = (2\pi\Delta_t)^{-L/2} e^{-\|u\|^2/2\Delta_t}$. Since $\nabla_x g_{\Delta_t}(x - e^{-t}a) = -\frac{x - e^{-t}a}{\Delta_t} g_{\Delta_t}(x - e^{-t}a)$ and, for x in a compact neighbourhood, $\|x - e^{-t}a\|\, g_{\Delta_t}(x - e^{-t}a) \le c_1 e^{-c_2\|a\|^2}$ uniformly, the integrand is dominated by an integrable function whenever P_0 has a finite first moment. Dominated convergence then licenses

$$\nabla_x P_t(x) = \int \mathrm{d}a\ P_0(a)\, \frac{e^{-t}a - x}{\Delta_t}\, g_{\Delta_t}(x - e^{-t}a).$$

Dividing by $P_t(x)$ and recognising $P_0(a)g_{\Delta_t}(x - e^{-t}a)/P_t(x) = P(a \mid x, t)$ gives (3). Nothing is approximated. In the empirical-Bayes literature this is Tweedie's formula [8–10].

Risk decomposition. For any u that is a function of x alone,

$$\mathbb{E}\big[\|a-u\|^2 \,\big|\, x\big] = \mathbb{E}\big[\|a-\langle a\rangle_{x,t}\|^2 \,\big|\, x\big] + \|u-\langle a\rangle_{x,t}\|^2, \tag{13}$$

by expanding $a - u = (a - \langle a\rangle_{x,t}) + (\langle a\rangle_{x,t} - u)$ and noting the cross term vanishes because $\langle a\rangle_{x,t} - u$ is x-measurable and $\mathbb{E}[a - \langle a\rangle_{x,t} \mid x] = 0$. Hence $\langle a\rangle_{x,t}$ minimises the conditional risk, with an irreducible floor given by the first term [49]. Optimality is with respect to squared error specifically; a different loss selects a different functional of the posterior. This is why every table in §7 reports error *against* $\langle a\rangle_{x,t}$ rather than raw risk: the floor is common to all estimators and subtracting it is what makes the comparison informative.

C What "exact" means here, in four parts

The body compresses this to two sentences; the distinctions matter enough to be kept apart somewhere, because collapsing any two of them is how a discretised number acquires the authority of an exact one.

(i) Sum-product on (4) is *exact at the level of functional messages*, as a consequence of the graph being a tree.

(ii) A message is a *function on* $\mathbb{R}$, and this is forced by the state space, not chosen. On a discrete graphical model a_i takes finitely many values and $\overrightarrow{m}_i$ is a vector of that length. Here a_i ranges over $\mathbb{R}$, so $\overrightarrow{m}_i$ assigns a number to every real a_i: it is an unnormalised density, the evidence from everything left of site i. Only when the transition and the likelihood are both Gaussian does that function stay inside a finite-dimensional family — two numbers, a mean and a variance, which is the Kalman filter. For a general φ it does not, and the integral in (5) produces a new shape at every site. An implementation must therefore represent a function. We use a truncated uniform grid $[-A, A]$ with N_{g} points, spacing h and composite trapezoidal weights, which defines a *finite-state model* for which the recursion is *exact up to floating-point arithmetic*.

(iii) That finite-state model *approximates* the continuous one, with an error that must be measured — §4 measures it.

(iv) Monte Carlo enters this note in exactly one place: when a statistic is computed from *generated samples*. Every other number here — posterior means, evidences, information matrices — comes from deterministic quadrature and carries no sampling error at all, only discretisation error. Where a quantity is averaged over random draws of a dataset, that is replicate spread and is labelled as such.

Three levels of exactness are involved and collapsing any two of them overstates the result, so each has its own name and none is abbreviated to "exact".

1. *Tree-exact functional BP.* The posterior factorises on a chain and the sum-product integrals are exact as statements about functions on $\mathbb{R}$. Nothing numerical has happened yet.
2. *Exact recursion for the discretised model.* Once grid and quadrature weights are fixed, forward–backward is exact — up to floating point — *for that finite-state model*. It is not exact for the continuous chain.
3. *Validated grid-quadrature reference.* The finite-state model approximates the continuous chain, with truncation and quadrature error that is measured rather than assumed (§4).

Accordingly the tables say "deterministic grid-BP reference under the true kernel", and the population statement is that the score *reduces to tree-exact functional inference and is evaluated numerically by controlled grid quadrature*. "Exact grid-BP" would collapse (ii) into (iii) and quietly promote a measured approximation into a guarantee.

D Sum-product on the posterior chain

Posterior factorisation. Bayes' rule applied to (2), with the likelihood factorising sitewise because the forward noise is independent across coordinates, gives (4). Each likelihood factor has degree one among latent variables, so attaching it to the prior's path adds a leaf and cannot create a cycle: the posterior factor graph is a tree. Conditioning usually creates dependence; it does not here, precisely because each observation is generated from a single latent variable. The construction fails the moment the forward noise is correlated across coordinates, since then $P(x \mid a)$ does not factorise, the likelihood becomes a factor touching several a_i, and the posterior graph acquires edges.

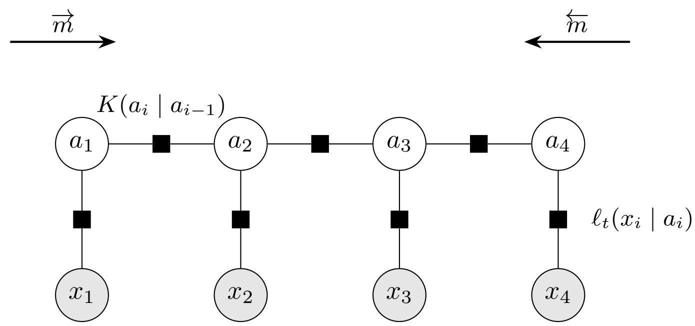

Figure 2: Factor graph of the posterior (4). Open circles are latent variables, shaded circles are observations, squares are factors. Each transition factor $K(a_i \mid a_{i-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor ℓ_t attached to a single latent variable. Conditioning therefore reweights node potentials without creating edges among the a_i, and the posterior graph remains a chain. Arrows show the two message sweeps (5).

Exactness on a tree. Let the graph be a tree and let a_i be a variable node. Removing a_i disconnects it into components, and every factor involves a_i together with variables of exactly one component — a factor touching two components would create a second path between them, hence a cycle. Grouping factors by component gives $P \propto \prod_c g_c(a_i, v^{(c)})$, so conditionally on a_i the components are independent. Unrolling the forward recursion (5),

$$\overrightarrow{m}_i(a_i) \propto \int \mathrm{d}a_{1:i-1}\ \mu(a_1) \prod_{j=2}^{i} K(a_j \mid a_{j-1}) \prod_{j=1}^{i-1} \ell_t(x_j \mid a_j),$$

which is the joint density of everything at or left of site i with the left variables integrated out and $\ell_t(x_i \mid a_i)$ *not* included; symmetrically for $\overleftarrow{m}_i$. The two are conditionally independent given a_i, so their product times the local evidence is proportional to the marginal. The pairwise statement is the same argument with the separator taken to be the edge.

Where the local factor goes. In (5) the local evidence at site $i-1$ is absorbed *before* the transition, so $\overrightarrow{m}_i$ carries the factors $\ell_t(x_1 \mid a_1), \ldots, \ell_t(x_{i-1} \mid a_{i-1})$ and $\overleftarrow{m}_i$ carries $\ell_t(x_{i+1} \mid a_{i+1}), \ldots, \ell_t(x_L \mid a_L)$, and each appears exactly once in the marginal. Absorbing it *after* would place $\ell_t(x_i \mid a_i)$ in both messages and square it: the belief would be sharper than the truth and every posterior variance too small. This produces plausible-looking output, which is why the convention is pinned by `tests/test_message_normalization.py` rather than left to a comment.

Evidence. Messages are renormalised at each step to prevent underflow; accumulating the discarded constants recovers $\log P_t(x)$ for the discretised model exactly. This is the objective that §5 maximises, and having it in closed form is what makes it possible to check a gradient against a finite difference of the thing being optimised. `src/bp_grid.py` $\to$ `grid_bp` returns it alongside the beliefs; `src/exact_scores.py` $\to$ `gaussian_log_evidence` is the closed form it is tested against.

Complexity at the functional level. Each message update integrates one variable against the kernel. Abstractly this is an integral operator applied to a function; represented on N_g points it is one $N_\mathrm{g} \times N_\mathrm{g}$ matrix–vector product, $O(N_\mathrm{g}^2)$. There are $2L$ updates, so one sequence costs $O(LN_\mathrm{g}^2)$ against $O(N_\mathrm{g}^L)$ for naive marginalisation. At $L = 32$, $N_\mathrm{g} = 401$ that is about 5×10^6 operations. Batching over sequences turns each update into a single GEMM, which is why the same code runs on a GPU without restructuring (`src/backend.py`, gated by `tests/test_backend_parity.py`).

E Covariance-stationary versus strictly stationary

Two notions of stationarity coincide for Gaussian processes and separate here, and the body relies on the weaker one, so the difference is worth stating.

A process is *strictly stationary* if the joint law of $(a_{i+1}, \ldots, a_{i+k})$ does not depend on i, for every k. It is *covariance-stationary* (or weakly, or second-order stationary) if only the first two moments are shift-invariant: $\mathbb{E}[a_i]$ constant and $\mathrm{Cov}(a_i, a_j)$ a function of $|i-j|$ alone. Strict stationarity implies the covariance version whenever second moments exist; the converse is false in general.

Why they coincide in the Gaussian case. A Gaussian law is determined by its mean vector and covariance matrix. If the mean is constant and the covariance depends only on $|i-j|$, then the joint law of any window $(a_{i+1}, \ldots, a_{i+k})$ is the Gaussian with that mean and that covariance block — and the block depends on i only through the differences $|j-j'|$, which are shift-invariant. So the joint law is shift-invariant, which is strict stationarity. The implication runs through "determined by two moments" and therefore fails for any family that is not.

What holds for our chain. With $q = 1-\rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly:

$$\mathrm{Var}(a_i) = \rho^2\,\mathrm{Var}(a_{i-1}) + q = \rho^2 + (1-\rho^2) = 1,$$

by induction, and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$, *whatever the shape of* φ. The chain is therefore covariance-stationary, and this is exactly the property the variance-matched family needs: Gaussian, Laplace, Student, uniform and bimodal innovations share the same second-order structure and differ only beyond it.

It is **not** strictly stationary for non-Gaussian φ. The invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is that of $\sum_{k\geq 0} \rho^k \varepsilon_{-k}$, an infinite convolution which is not Gaussian, so drawing $a_1 \sim \mathcal{N}(0,1)$ starts the chain off its invariant law. The marginal of a_i then drifts towards it over a burn-in of order $1/\log(1/\rho) \approx 6$ sites at $\rho = 0.85$. This does not affect the comparisons in the body, because every family shares the same a_1 law and the same covariance and so still differs only beyond second moments — but $q = 1-\rho^2$ alone does not deliver strict stationarity, and we do not claim it does.

F The Gaussian model and the second-order baseline

For $\varphi = \mathcal{N}(0, q)$ with $q = 1-\rho^2$ and $a_1 \sim \mathcal{N}(0,1)$, the vector a is jointly Gaussian with $(\Sigma_0)_{ij} = \rho^{|i-j|}$. Expanding $P_0(a) \propto \exp[-\frac{1}{2q}\sum_{i\geq 2}(a_i - \rho a_{i-1})^2 - \frac{1}{2}a_1^2]$ produces only a_i^2 and $a_i a_{i-1}$ terms, so Σ_0^{-1} is tridiagonal with $(\Sigma_0^{-1})_{ii} = (1+\rho^2)/q$ in the interior, $1/q$ at the two ends, and $-\rho/q$ on the off-diagonals. Adding the likelihood contributes $(e^{-2t}/\Delta_t)I$, so the posterior precision is tridiagonal at every t: Markovianity is a statement about the precision, while the covariance Σ_0 is dense.

Since $x = e^{-t}a + \sqrt{\Delta_t}z$ with z independent, (a, x) is jointly Gaussian with $\mathrm{Cov}(a, x) = e^{-t}\Sigma_0$ and $\mathrm{Cov}(x) = e^{-2t}\Sigma_0 + \Delta_t I =: \Sigma_t$, whence $\mathbb{E}[a \mid x] = e^{-t}\Sigma_0\Sigma_t^{-1}x$ and, by (3), $S(x,t) = -\Sigma_t^{-1}x$. For zero-mean jointly Gaussian variables the conditional expectation is linear, so it coincides with the LMMSE estimator [50]. That supplies the covariance identity the proposition's proof appeals to; the proof itself, an induction over the forward and backward recursions establishing that every projected message equals the Gaussian chain's, follows.

Remark 2 (On the hypotheses)**.** *Four are load-bearing and worth separating.* Finite second moments *are what make* Π_{G} *defined at all.* Zero mean *is not cosmetic: the general LMMSE carries* $\mathbb{E}[\mathbf{a}] + \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}(\mathbf{x} - \mathbb{E}[\mathbf{x}])$*, and* (8) *is the centred special case.* $\rho \neq 0$ *is required because the backward map of the Gaussian closure recursion divides by* ρ*; at* $\rho = 0$ *the chain has no edges, the sites are independent, and the claim holds trivially and separately, each site's posterior mean being the scalar Gaussian* $e^{-t}\boldsymbol{\Sigma}_0(e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t)^{-1}x_i = e^t\boldsymbol{\Sigma}_0(\boldsymbol{\Sigma}_0 + \Delta_t e^{2t})^{-1}x_i$. Where the projection is applied *matters because* Π_{G} *does not commute with multiplication by the likelihood; Proposition 1 fixes the order, and a different order is a different algorithm.*

The statement is about exact message functions on $\mathbb{R}$ *and an analytic projection operator.* ***The grid plays no part in it.*** *Discretisation enters only when the* unrestricted *non-Gaussian messages are represented numerically, which is what the recursion is compared against; conflating the two would turn a theorem about a closure into a claim about quadrature.*

Proof. Write $\mu_i^{\rightarrow}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message coincides with the corresponding exact message of the covariance-matched Gaussian chain — the chain with the same ρ and innovation variance q but Gaussian innovations.

Base. The initial message is Gaussian by hypothesis, or is projected to the Gaussian with the same first two moments; either way it equals the Gaussian chain's initial message, which has those same moments by construction.

Step. Suppose $\mu_{i-1}^{\rightarrow}$ agrees with the Gaussian chain's message, so it is Gaussian with some mean m and variance v. Multiplying by ℓ_{i-1} multiplies two Gaussian densities and yields a Gaussian, with mean and variance given by the usual precision-weighted combination — identical in both chains, since the likelihood is the same object. Taking the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ convolves with φ and scales by ρ, which maps $(m, v) \mapsto (\rho m,\, \rho^2 v + q)$: the mean and variance of a convolution depend on φ *only* through its mean and variance, so this is the same map for every innovation with moments $(0, q)$, and in particular the same as for the Gaussian chain. In the Gaussian chain the convolution of two Gaussians is already Gaussian, so no projection occurs there; here Π_{G} is applied, and since the pre-projection function has exactly the moments just computed, the projection returns precisely the Gaussian chain's message.

The backward recursion runs the same argument with the transition map inverted, $(m, v) \mapsto (m/\rho,\, (v + q)/\rho^2)$, which is where $\rho \neq 0$ is used. Its base case needs one word of care. The terminal backward message is $r_n \equiv 1$, which is not a normalisable density and so is not in the domain of Π_{G}; the induction therefore starts one step earlier, at the terminal *unary likelihood* ℓ_n, which is Gaussian and normalisable. Every subsequent backward message is a likelihood-weighted transition integral of a normalisable Gaussian and stays in the domain. Nothing is lost by starting there, because $r_n \equiv 1$ contributes no factor: the first object the recursion actually projects is ℓ_n.

Hence every projected belief equals the corresponding Gaussian-chain belief, and the returned posterior mean is the Gaussian chain's posterior mean. For jointly Gaussian zero-mean $(\mathbf{a}, \mathbf{x})$ the conditional mean is linear in $\mathbf{x}$ and therefore coincides with the best linear estimator, which is (8): since $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\mathbf{z}$ independent, $\mathrm{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t}\boldsymbol{\Sigma}_0$ and $\mathrm{Cov}(\mathbf{x}) = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$. Since the recursion never read φ beyond $(0, q)$, the same estimator is returned for every innovation law with those moments. $\square$

Consequently, on this model, the *message-representation error* and the *model error* coincide: projecting messages onto Gaussians and replacing the prior by its covariance-matched Gaussian give the same estimator. This is a property of single-Gaussian closure with linear transitions, not of Gaussian approximations in general, and the two separate for richer message families.

G Numerical representation

Messages are represented on a uniform grid $u_1, \ldots, u_{N_{\mathrm{g}}}$ spanning $[-A, A]$ with spacing $h = 2A/(N_{\mathrm{g}} - 1)$ and composite trapezoidal weights $w_k = h$ for interior points, $h/2$ at the two ends.

The recursions are *scaled linear-domain* forward–backward, not log-semiring belief propagation. The transition matrix and the unary likelihoods are built in logs — the likelihood with a per-site maximum subtracted before exponentiating — and are then exponentiated once, so each message update is an ordinary $(N_{\mathrm{g}}, N_{\mathrm{g}})$ matrix product followed by renormalisation to unit quadrature mass. The removed constants are accumulated and restored in the evidence, which is therefore exact. This is the standard scaled recursion and it is what keeps the *total* mass of every message at unity, so no message vanishes as a whole.

It does not, however, guarantee that no individual entry underflows: a single likelihood or transition value far in a tail can round to zero while the message it belongs to remains well-normalised. The recursion is scaled, not logarithmic, so this is a real possibility and is measured below rather than excluded by construction. The 10^{-14} agreement reported below is correspondingly evidence at the Gaussian configuration tested, not a uniform bound over innovation families, noise levels and fitted kernels.

How far from a bound, measured. Sweeping $N_\mathrm{g} \in \{201, 401, 801\}$ and $A \in \{6, 8, 10\}$ against a finer reference ($N_\mathrm{g} = 1601$, $A = 12$), over all twelve noise levels and four innovation families, with observations drawn from the prior and again conditioned into the tail at $\sim$3 standard deviations:

worst relative score error at $N_\mathrm{g} = 401,\ A = 8$	Gaussian	bimodal	Laplace	Student
observations drawn from the prior	2×10^{-13}	1×10^{-15}	1×10^{-5}	$\mathbf{9.9 \times 10^{-4}}$
observations conditioned into the tail	2×10^{-7}	1×10^{-11}	$\mathbf{1.7 \times 10^{-2}}$	8.3×10^{-3}

The 10^{-14} figure is the *best* cell in this table, not a bound on it. Under ordinary draws the production grid resolves the whole schedule to 10^{-3} or better across every family, which is the regime all reported results are in and is the threshold the generator enforces. Under displaced observations the two heavy-tailed families lose two further orders. Partial underflow is real rather than hypothetical: up to 0.27% of likelihood entries round to exactly zero at the production configuration, rising to 3.8% at $N_\mathrm{g} = 201$ — always under displacement, never under ordinary draws.

What the stress row is, exactly. It is a three-standard-deviation *displacement* of the whole chain, not conditioning on a tail event of the prior. A translate and a conditional law are different objects, and it is the translate that is computed: a legitimate stress test, named as one.

Boundary occupancy, measured inside the recursion. The quantity that matters is how much mass the objects BP actually propagates place at the edge of the domain, so it is measured on those objects rather than on a proxy assembled from the likelihood and the transition matrix. The worst forward-message edge occupancy over the schedule is $6.6e-05$ and the worst node-belief edge occupancy is $2.9e-03$, both over 60 production cells. The worst per-site log-evidence shift against the reference resolution is $1.0e-05$ nats.

Fitted kernels, the hardest case for the grid. The families above are four analytic priors, but the kernels the reported runs actually use are *fitted* mixtures, which can place a component far narrower than any smooth law. Since the narrowest component relative to the mesh is precisely what the grid must resolve, this is the case most likely to fail and is tested directly. Adding a fitted mixture with a component two grid cells wide changes the picture in one place and not the other. Under ordinary draws it is unremarkable and the schedule still resolves to $9.9e-04$. Under three-standard-deviation displacement at the narrowest domain tried ($A = 6$) it is the worst cell in the entire sweep, with 96% of node-belief mass against the boundary: a narrow component displaced toward the edge of a too-small window is not resolved at all. That configuration is outside the production setting, and it marks where the method stops working rather than a defect in the reported numbers — but it is the failure mode to check first if the domain is ever narrowed.

What this does and does not certify. It is a self-convergence stress test on fresh synthetic draws. It establishes that errors on representative ordinary draws are below 10^{-3}. It does *not* establish that no reported number moves: that would require re-evaluating the actual held-out observations and fitted kernels at a finer resolution — a different experiment, and one not run here. Section L records the residual limitation.

Two error sources. *Truncation*: probability mass outside $[-A, A]$ is discarded rather than transported, an error controlled by A and essentially independent of N_g. *Quadrature*: the trapezoidal rule is second order in h for integrands with bounded second derivative, an error controlled by h at fixed A. Figure 3 measures them separately, using the forward-message boundary mass for the first and the interior column-mass residual of the transition matrix for the second; the interior restriction $|u| \leq A/2$ removes columns whose transition density has a tail outside the grid, which is what makes the second diagnostic measure quadrature alone.

The boundary mass is the maximum over sites of a *sampled* trajectory, so it is a statistic and not a bound, and it is reported over 256 chains per cell rather than from one draw. At $N_\mathrm{g} = 401$, taking the worst case over the twelve-level schedule,

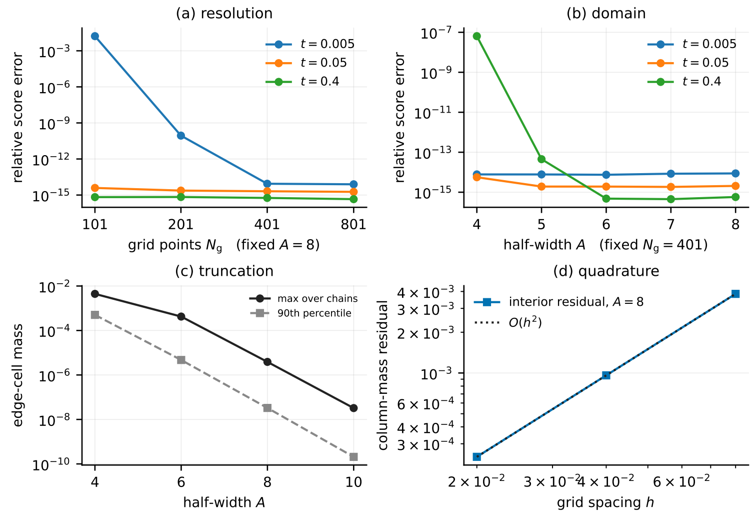

Figure 3: Grid and domain control, Laplace chain. (a) Relative score error against a high-resolution reference at fixed $A = 8$; (b) the same at fixed $N_g = 401$ as the domain grows. Both saturate at the floating-point floor $\sim 10^{-14}$, so beyond that the curves measure arithmetic, not discretisation. (c) Forward-message boundary mass against A — the truncation diagnostic — as the worst case and the 90th percentile over 256 chains, which differ by roughly two orders of magnitude and are therefore both drawn. (d) Interior column-mass residual against spacing at $A = 8$ — the quadrature diagnostic — with an $O(h^2)$ reference. The two diagnostics do different jobs and neither alone certifies convergence.

A	4	6	8	10
worst chain	2.3×10^{-3}	1.6×10^{-4}	1.2×10^{-6}	3.3×10^{-8}
90th percentile	2.7×10^{-4}	2.5×10^{-6}	1.5×10^{-8}	7.8×10^{-11}
median	3.8×10^{-5}	3.8×10^{-7}	3.0×10^{-9}	1.8×10^{-11}

The spread between the rows is the point: at the working $A = 8$ the worst chain carries 84 times the boundary mass of the 90th percentile, so a single trajectory's value — or an upper quantile quoted as though it were a bound — understates the tail by close to two orders of magnitude. The interior residual falls by a factor of four per halving of h, i.e. $O(h^2)$.

At $A = 4$ the interior residual saturates at $\sim 10^{-3}$ instead of continuing to fall, because at that domain even the "interior" columns have truncated tails — the floor is the truncation error leaking into the quadrature diagnostic. This is the reason the two diagnostics have to be read together and neither alone certifies convergence.

What is not claimed. A grid-resolution sweep at fixed A does not rule out truncation error, and a domain sweep at fixed N_g does not rule out quadrature error. We report both axes and the two diagnostics; we do not claim continuum convergence, only that at $N_g = 401$, $A = 8$ both diagnostics sit far below every effect the paper reports. For the Laplace family, whose innovation density has a discontinuous derivative at the origin, we claim second order and no better.

H Learning procedure

Objective. Maximum marginal likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(x^\mu)$, with the complete-data objective $\log P_\theta(a, x) = \log \mu(a_1) + \sum_{i\geq 2} \log K_\theta(a_i \mid a_{i-1}) + \sum_i \log \ell_t(x_i \mid a_i)$. Only the middle term depends on θ, which is why $\boldsymbol{\Xi}$ of (9) is sufficient and why $\mathcal{Q}(\theta \mid \theta') = \langle \boldsymbol{\Xi}(\theta'), \log K_\theta \rangle$ up to θ-free constants.

E-step. One forward and one backward sweep per sequence, then a rank-one accumulation of the pairwise statistics: writing $\ell^{(i)}$ for the vector $\ell_t(x_i \mid \cdot)$ on the grid, and with $f_i = w \odot \overrightarrow{m}_i \odot \ell^{(i)}$ and $g_i = w \odot \overleftarrow{m}_i \odot \ell^{(i)}$, the pairwise belief on edge $(i, i+1)$ is $f_i(j)\,K[k,j]\,g_{i+1}(k)/Z_i$ with $Z_i = g_{i+1}^\top K f_i$, and $\Xi = \left(\sum_{\mu,i} g_{i+1}(f_i)^\top / Z_i\right) \odot K$. Ξ sums to the number of edges in the dataset, which is asserted in the tests. Exact for the discretised model, in the sense of Appendix C(ii). Implementation `src/em.py` $\to$ `e_step`.

M-step, by kernel.

- *Gaussian*, $\theta = (\rho, q)$: closed-form maximiser, the belief-weighted Yule–Walker equations $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$. Exact maximisation, hence exact EM for that family.
- *Laplace*, $\theta = (\rho, b)$: closed form despite non-differentiability. Profiling b out of $\mathcal{Q}$ turns the problem into minimising the belief-weighted mean absolute residual, whose exact solution is a weighted median. Exact maximisation.
- *Mixture* (12): the complete-data problem carries a second latent variable, so the M-step is itself an inner EM over the mixture label with Ξ held fixed. We alternate the (π, ν, s^2) block and the ρ block, both closed form, for at most **sixteen** inner sweeps, stopping early when the per-edge gain in $\mathcal{Q}$ drops below 10^{-8}. This increases $\mathcal{Q}$ without maximising it: **generalised EM / ECM** [35], not exact EM. Sixteen is chosen by measurement: at four sweeps the fitted innovation excess kurtosis is 28% below its value at sixteen whenever the outer loop is also cut short, while ρ moves in the fourth decimal — the same shape-versus-correlation rate asymmetry the outer loop shows, one level down. Since shape is the coordinate every non-Gaussian claim depends on, the budget is reported here rather than left as a default. Because each block is a conditional *maximiser*, $\mathcal{Q}$ cannot decrease along an inner sweep; the implementation raises rather than stopping if it ever does, and across the certified fits it never has.
- *Mixture-density network*: one backtracked Adam step, accepted only if $\mathcal{Q}$ increased, halving the step up to five times otherwise and standing still if no ascent is found. Also generalised EM.

Guarantees, initialisation, stopping. Generalised EM guarantees monotone ascent of $\mathcal{L}$ [33, 34]; convergence is to a stationary point, not necessarily a global maximum. Mixture components are initialised spread over $[-\sqrt{\mathrm{Var}}, \sqrt{\mathrm{Var}}]$ with random scales; ρ is initialised away from its truth in every experiment — at 0.3 in most, and over $\{0, 0.3, 0.6, -0.4\}$ in the initialisation sweep of `exp_18`. The truth is 0.85 in `exp_02`, `exp_16`, `exp_18` and `exp_22`, and 0.8 in `exp_06`, `exp_07` and `exp_08`; every table and figure states the value it used. Iteration stops when the relative increase of $\mathcal{L}$ falls below 10^{-9}, capped at 50–120 iterations depending on the experiment. Variances are floored at 10^{-6} to prevent component collapse; no other regularisation is applied, and the component means are unconstrained (§5). Monotonicity is recorded per run as the largest decrease across iterations and is zero in every run reported here.

I Convergence rates, clean against channel

Two slownesses, measured separately, because they are independent and easily conflated.

Optimisation: coordinates of the same kernel settle at very different rates. Running EM to 800 updates at $M = 512$, $C = 8$ and recording the whole parameter trace, we measure for each coordinate the last update at which it was still further than a fixed relative tolerance from its final value. Over sixteen seeds the median is 80 updates for ρ (10^{-3} tolerance), 68 for the innovation variance (10^{-2}), and 229 for the excess kurtosis (2×10^{-2}), the last ranging up to 638. The shape is thus roughly three times slower to settle than the correlation, and monitoring ρ — the natural thing to plot — shows a converged-looking trace over a kernel that is not. This is why the stopping rule may not be read off ρ, and why an iteration budget must be justified against the *slowest* coordinate.

Statistics: fitting through the channel costs rate, not just constant. Comparing maximum likelihood on clean transition pairs against the same estimate obtained through the noising channel, at sixteen replicates over $M = 64$ to 1024, the fitted $\log$–$\log$ slopes are

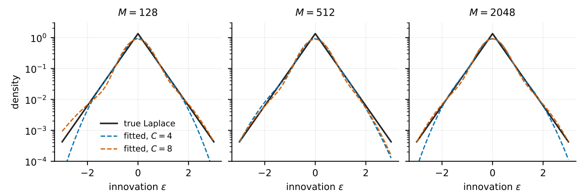

Figure 4: The fitted innovation density against the true Laplace density, on a log scale so the tails are visible. What is recovered here is a density, not a parameter: the mixture is a representation of it.

	ρ	innovation variance
clean transition pairs	-0.52	-0.50
through the channel	-0.44	-0.36

The clean arm sits on the parametric $M^{-1/2}$ to two decimal places on both coordinates, which is the check that the estimator itself is not the bottleneck; the noised arm is slower, and more so on the shape than on the correlation. Note that the clean arm is *not* flat in M — its error falls by a factor of 3.4 on ρ and 4.3 on the variance across the range. This is worth stating explicitly: a flat clean arm would indicate a discretisation floor, and at three replicates the curve is noisy enough to look flat when it is not.

J The fitted rates in the efficiency comparison

The body reports fitted $\log$–$\log$ slopes of relative denoising error against M. They are fitted *per seed* and then averaged, not fitted to the seed-averaged curve: cells within a seed share one training set and one set of fitted models, so the two differ, and only the first keeps the pairing that makes the arms' difference testable. Over $M = 32$ to 4096 the network gives -0.259 ± 0.002 and EM–BP -0.378 ± 0.012, a paired difference of 0.119 ± 0.011 ($t = 10.7$ on $16 - 1$ degrees of freedom). The sign test is the one to trust at this sample size, and it is unanimous: EM–BP is steeper in all 16 seeds. Restricting to $M \geq 256$ or to $M \leq 512$ leaves the difference at 0.13 and 0.10 respectively, so it is not carried by either end of the range.

Why censoring does not explain it. Both arms select their largest allowed budget in a growing fraction of cells as M grows — the network from 6% at $M = 32$ to 42% at 4096, EM–BP from 12% to 61%. By the rule used throughout this work, two arms with different cap-hit rates are being compared on their budgets, so the fitted slopes are uninterpretable unless the direction of the effect is known. It is measured. Tripling both caps at $M = 2048$, on the same seeds and validation bundles, lowers the network's error by 0.35% and EM–BP's by 4.53% — an order of magnitude apart. EM–BP is the arm being held back, and it is also the arm whose cap-hit rate rises faster with M, so the censoring inflates its error more at large M than at small and *flattens* its fitted curve. Removing the censoring would steepen EM–BP's slope and widen the difference. The reported value is a lower bound.

Two further checks point the same way. Fitting the slope between $M = 2048$ and $M = 4096$ using only the raised-budget runs at both ends — so that neither endpoint is at the certified cap — gives a difference of 0.26 ± 0.12, larger than the full-range figure, though a two-point slope on 16 seeds is individually weak. And the fit is generated rather than transcribed: the generator refuses to emit if the budget calibration ever shows the network as the more censored arm, since that is the single assumption the claim rests on which the fitted slopes do not themselves display.

What the slopes are not. Neither matches a rate we can derive. EM–BP's -0.378 is well short of the $M^{-1/2}$ that Appendix K and the Rung 1 measurement obtain for q alone, which is consistent with a pointwise error inheriting the slower of its two coordinates rather than the faster — ρ falls at -0.27 over the same range — but we have not established that, and we do not build on it. The claim in the body is the qualitative one.

K Experimental protocol

Remark 3 (Discrete and continuous are separate claims)**.** *Proposition 2 is a statement about the* finite-state model on the grid*. There the interchange of ∇_θ with a finite sum is unconditional, and the returned gradient is exact for the discretised likelihood up to floating point — which is what the* 10^{-9} *finite-difference agreement of Appendix K measures.*

The corresponding continuum identity, $\nabla_\theta \mathcal{L} = \iint \xi_\theta(a, a') \nabla_\theta \log K_\theta(a' \mid a) \, \mathrm{d}a \, \mathrm{d}a'$, also holds, but it needs hypotheses the discrete statement does not: $K_\theta(a' \mid a)$ differentiable in θ, support not depending on θ, and an integrable dominating function for $\partial_\theta K_\theta$ so that differentiation and integration may be exchanged.

Gaussian and Gaussian-mixture innovations satisfy the hypotheses as written*: both are smooth in θ with full support. Laplace does not, and saying so matters because Laplace is the innovation law the reported runs use. Its density $\propto \exp(-|a' - \rho a|/b)$ has a kink at $a' = \rho a$, so $\partial_\rho K_\rho$ fails to exist on that set and the proposition's smoothness hypothesis is violated there. The identity survives, but by a different argument: the kink set is Lebesgue-null, the derivative exists almost everywhere, and the difference quotients are dominated locally uniformly in ρ, so dominated convergence still licenses the exchange. The discrete statement is untouched either way — on the grid the kink is a finite set of entries and the sum is differentiated term by term — and it is the discrete one the reported numbers certify.*

What Ξ is sufficient for. The complete-data log-likelihood is a sum over edges, so the θ-dependent part of $\mathcal{Q}(\theta \mid \theta^{(r)})$ sees the data only through Ξ. But Ξ is itself computed at $\theta^{(r)}$ and rebuilt every iteration, so the claim concerns one maximisation step, not the data being compressible once. It also requires a homogeneous kernel shared across sites and sequences, a fixed grid, an initial law excluded from θ, and θ-free likelihood factors — the last licensing observations at several noise levels to add into one Ξ. Given those, maximisation costs $O(PN_{\mathrm{g}}^2)$ for P parameters, independently of M, while forming Ξ costs $O(MLN_{\mathrm{g}}^2)$ per iteration. "One matrix" is therefore a statement about sufficiency, not about inference being cheap.

The reported numbers come from one frozen configuration, defined in a single file and imported by the experiments that produce them: Laplace chain unless stated, $\rho = 0.85$, $L = 32$, $N_{\mathrm{g}} = 401$, $A = 8$, twelve log-spaced noise levels on $[0.05, 3.0]$, sixteen replicates, and 256 held-out sequences on a disjoint seed.

Two qualifications limit how much the frozen file guarantees on its own. A declared field is not necessarily a *read* field: importing a configuration does not make the code consult it, and the provenance stamp compares the configuration against itself, so it cannot see a setting nobody reads. Two such fields exist today and are listed as defects in `tests/test_frozen_config_is_read.py`, which fails the build if a third appears. Separately, experiments carrying their own local settings — chain length, noise grid, architecture sweeps — are not described by this file at all, so their outputs additionally record a digest of the settings actually resolved at run time.

What Table 1 covers. Four qualifications, kept here rather than in the caption. Both arms select their optimisation budget on a disjoint validation bundle — the network its parameterisation *and* training length, EM–BP its iteration count — and that choice agrees with the test-set optimum in 81.1% and 77.6% of cells respectively, so the protocol is symmetric and the selection is close to an oracle without being one. The $M = 4096$ row uses a raised budget, the caps having been set for $M \leq 2048$; rerunning $M = 2048$ at that budget on the same seeds moves its ratio by -0.16 ± 0.18, so the caps are selected without binding. 16 of $1{,}536$ cells are excluded by the resolution gate, because the narrowest fitted mixture component is under two grid cells wide; including them changes no ratio by more than 0.08. And the network is beaten in all but 2 of the reported cells, both at the smallest sizes — Appendix P resolves where the advantage is largest.

An excluded row, and why the neighbouring one stays. No $M = 8192$ row is reported, although one was run. Its parameter file records a source tree with 207 uncommitted paths, and intersecting that list with the experiment's fifteen-file import closure leaves three that the run itself executes: the experiment, the frozen configuration, and the EM implementation. The program that produced those numbers is not recoverable, so the row is excluded from Table 1 and from every figure drawn on the same grid; it would otherwise have set the top of the quoted ratio range.

The $M = 4096$ rows are kept, and the distinction is not special pleading. Their tree carried eight uncommitted paths and the same intersection is *empty* — all eight are wavelet and image-metric work, none of it reachable from this experiment — so no edit in that tree could have moved these numbers. “The tree was dirty” and “a file this run executes was dirty” are different facts, and only the second is disqualifying. The check is `tools/provenance_gate.py`, which computes the closure statically and reports the intersection per run rather than refusing every dirty tree alike.

One view per chain. Each clean chain is noised at *one* level, the chains being split evenly across the twelve levels, so a training set of M chains yields exactly M observations and the likelihood is a genuine one-view marginal likelihood rather than a composite objective over repeated views of the same latent sequence. This costs resolution at the small end — at $M = 32$ a level carries two or three chains — and it is what makes M a count of independent latent sequences, so that the horizontal axis of Table 1 is a sample size and not a count of noised views.

The stopping rule, and what actually gates a reported number. Two mechanisms are in play and only the second determines any headline result.

`fit_em` stops when the signed log-evidence gain per transition falls in $[0, 10^{-9}]$ — signed, so a decrease is recorded as a defect rather than accepted as convergence — subject to an iteration cap supplied by the caller, after which the run is labelled *censored* rather than converged. This is a diagnostic: it says whether a fit ran to a flat likelihood, and nothing more.

The headline estimator does not rely on it. It selects an iteration count on the disjoint validation bundle, from the grid listed below, exactly as the network arm selects its training length — so the reported model is chosen by held-out risk, not by a convergence test, and a censored fit is not thereby disqualified.

That distinction carries real weight here, because the coordinates converge at very different rates: at $\rho = 0.85$, $M = 512$, $C = 8$ the autoregressive coefficient is flat from iteration $\sim$25 while the fitted excess kurtosis reads 0.84 at 30 iterations, 1.85 at 60, 2.30 at 120 and 2.29 at 400. A likelihood- or ρ-based stopping rule would certify a kernel still missing a third of the higher-order structure. Validation selection is what guards against that, and it is the guard every reported number passes through. The configuration file also declares a shape-based threshold, $|\Delta\kappa_4| < 10^{-3}$; nothing reads it, it is one of the two unread fields recorded above, and no reported number was produced by it.

chain	Laplace AR(1), $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0, 1)$, $L = 32$
other families	Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$)
	all variance-matched to q, so $\mathrm{Cov}(a_i, a_j) = \rho^{\lvert i-j \rvert}$ exactly
grid	$N_\mathrm{g} = 401$, $A = 8$ ($h = 0.04$); refinements $N_\mathrm{g} \in \{801, 1601\}$
noise levels	twelve log-spaced $t \in [0.05, 3.0]$, for training and pointwise evaluation
budgets	$M \in \{32, 64, 128, 256, 512, 1024, 2048\}$, one noise view per chain
test set	256 held-out sequences, seed `exp07-test`, disjoint from all training seeds
validation	a second disjoint bundle, used to select both arms' budget (below)
replicates	sixteen; the replicate index is mixed into every training seed, so replicates
	vary the training set *and* the initialisation; the test set is common to all
EM–BP	mixture-innovation kernel (12), $C = 8$; $3C = 24$ free parameters
	ρ initialised at 0.3; iteration count selected on validation from
	$\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400\}$; relative tolerance 10^{-9}
global MLP	fully connected, 25,248 parameters, denoising score matching, Adam
	training length selected on validation from $\{0.5, 1, 2, 3.5, 6, 10, 14, 20\} \times 10^3$ steps
	both parameterisations (ε and a) trained, and the choice between them
	is also made on validation
local CNN	weight-shared one-dimensional convolution, radius r, receptive field $2r + 1$
capacity control	24 configurations, 3.2k–297k parameters
hardware	CPU: single-threaded, `medium_cpu` partition, Bocconi HPC
	GPU: CUDA 12.4, CuPy 13.6, gated by a CPU/GPU parity test before any sweep

Selection rules, stated plainly. Every tunable quantity that either arm is allowed to choose is chosen on a validation bundle disjoint from both the training set and the test set, and the two arms are given the same privilege. The network selects its parameterisation (ε or a) *and* its training length; EM–BP selects its iteration count. Neither selection touches the test set. The selected choice coincides with the test-set optimum in 82.2% of cells for the network and 79.5% for EM–BP, and Table 1 reports the validation-selected numbers, not those optima.

The symmetry matters because the training length is a bias–variance knob for both arms, not a convergence detail: selecting one arm's budget while pinning the other's is not a neutral choice. Measured on this run, giving the network the same early-stopping privilege costs between 5% and 36% of the ratio, the larger corrections at the smaller sample sizes. The one selection that remains a fixed configuration value rather than a validation choice is the convolution's receptive field; the numbers in §7 that depend on it are labelled exploratory.

L Scope conditions

The initial law is held fixed at the one the generating process uses, so nothing is misspecified there and nothing shows it could be recovered. The linear form of (12) is assumed: what is free is a scalar and a density, not an arbitrary bivariate transition. The chain is covariance-stationary rather than strictly stationary. Identifiability is a theorem at level 1 and conditional at level 2. The implementation is a discretisation whose error, for the Laplace family, is second order rather than spectral. Inference costs $O(LN_\mathrm{g}^2)$ per sequence, far above a network forward pass, so the estimator's advantage is in data and not in wall-clock. And the comparison is asymmetric in supplied information by construction (Section 7): it measures what the Markov assumption buys where that assumption holds, and says nothing about a setting where it does not.

M Reproducibility map

claim	script (`experiments/`)	output (`outputs/`)
grid and domain convergence	`exp_01_grid_validation`	`exp_01_.../grid_heatmap.csv`
boundary and quadrature diagnostics	`exp_18_... --parts boundary`	`exp_18/boundary.csv`
closure error vs t	`exp_02_laplace_gaussian_...`	`exp_02_.../laplace_summary.csv`
closure error by family	`exp_03_nongaussian_...`	`exp_03_.../innovation_sweep.csv`
EM monotonicity, ρ recovery	`exp_18_... --parts emtrace`	`exp_18/em_trace.csv`
learned innovation density	`exp_18_... --parts density`	`exp_18/innovation_density.csv`
parameter-recovery rate	`exp_06_em_parameter_...`	`exp_06_.../em_rate.csv`
Fisher information vs t	`exp_06_em_parameter_...`	`exp_06_.../price_of_noising.csv`
Fisher identity vs finite diff.	`exp_08_gradient_vs_...`	`exp_08_.../gradient_vs_exact.csv`
sample efficiency (§7)	`exp_07_em_vs_score_...`	`replicates/merged_summary.csv`
network capacity control	`exp_07_em_vs_score_...`	`exp_07_.../capacity.csv`
inference and training cost	`exp_07_em_vs_score_...`	`exp_07_.../{inference,training}_cost.csv`
receptive field (exploratory)	`exp_12_receptive_field`	`exp_12_.../efficiency_three_way.csv`
sampler step convergence	`exp_16_... MODE=calibrate`	`exp_16/calibrate_steps*/steps.csv`
generation by budget	`exp_16_... MODE=generate`	`exp_16/gen_n*_seed*/generation.csv`
generation by family	`exp_16_sampling_validation`	`exp_16/family_*/generation.csv`
capacity: generation	`exp_16_sampling_validation`	`exp_16/components_C*/generation.csv`
capacity: pointwise	`exp_16_sampling_validation`	`exp_16/cpoint_C*/pointwise.csv`
discrete-alphabet variant	`exp_10_discrete_alphabet`	`exp_10_.../discrete_*.csv`

Every figure in this note is produced by `paper/figures/make_figures.py`, which reads the outputs above and fails loudly if one is missing; no number in any figure is entered by hand. Exact commands, environment and HPC job configuration are in `REPRODUCIBILITY.md`, and the claim-level audit behind this revision is `REVISION_AUDIT.md`.

N The structured baseline

The headline comparison of Section 7 uses a fully connected network, which carries no locality prior and no weight sharing. Separating that from the structural assumption requires a weight-shared window predictor with radius and parameterisation selected on a validation bundle disjoint from test, and EM–BP refitted on the same chains — at the headline protocol, with every site scored, deployed and provenance-gated through `hpc/deploy_clean.sh`. `experiments/exp_31_structured_baseline.py` is that measurement: it replaces an earlier exploratory run whose provenance did not verify and whose protocol differed from the headline table's in four ways (`outputs/README_exp12_scaled.md` records the detail; no number from that run is cited here or in the main text).

Which architecture, and under what search. Three candidates were screened on 4 development seeds, disjoint from the 16 used below: the weight-shared window head, a dilated convolutional stack, and a bidirectional message-passing network, the last two able in principle to propagate across the whole chain. Each ran over its own hyperparameter grid, with a shared learning-rate grid, both output parameterisations and a common checkpoint ladder (500 to 64000 steps), 32 configurations in all, selected on mean risk over development seeds on the $bulk$ region. **The window head won**, at radius 4, width 64.

Table 2: The architecture screen that selected Table 3's baseline. Each candidate architecture was run over its own hyperparameter grid on 4 development seeds (disjoint from the confirmatory seeds), at $M \in \{32, 2048\}$, with a shared learning-rate grid, both output parameterisations, and a shared checkpoint ladder; the reported risk is the mean over development seeds on the interior slice used for selection. $\star$ marks the winner carried into the confirmatory run. The dilated stack carries an order of magnitude more parameters than the winner and finishes last, so the outcome is not an artefact of one arm being given more capacity than another.

architecture	configurations	parameters	selection risk
weight-shared window head $\star$	16	4.8k–21.4k	0.137
dilated convolutional stack	8	66.4k–264k	0.205
bidirectional message passing	8	9.0k–34.3k	0.172

This is a model-selection fact under this search space and protocol, not evidence that long-range propagation is intrinsically unnecessary or that radius 4 is optimal in the population. It does rule out the cheapest objection: the dilated stack carries $66.4k - -264k$ parameters against the window head's $4.8k - -21.4k$ and placed last, so the long-range arms did not lose for want of capacity. The confirmatory comparison below is therefore against the strongest baseline *this screen found*, not the strongest conceivable one, and every claim that follows is scoped to that.

Table 3: EM–BP against the screened structure-aware baseline, at the headline protocol throughout. The baseline is a weight-shared window head whose architecture and hyperparameters were fixed by the screen of Table 2 on development seeds; only the checkpoint is chosen here, on a validation bundle disjoint from test, and EM–BP's iteration count is chosen the same way. Risks are all-site and averaged over the 16 seeds; the ratio is risk(window)/risk(EM–BP) formed within each seed and then averaged, $\pm$ one s.e. *All* scores every site, and is the headline; *centre* is a single-site diagnostic. A third region, the predeclared interior slice, is scored in the stored output and used for architecture selection but not reported here.

	risk, all sites		ratio	
M	EM–BP	window	all sites	centre site
32	0.1005	0.2006	2.34 ± 0.26	2.04 ± 0.24
128	0.0652	0.1594	2.82 ± 0.30	2.35 ± 0.26
512	0.0394	0.1353	3.58 ± 0.18	3.25 ± 0.18
2048	0.0224	0.1267	6.14 ± 0.45	5.43 ± 0.39

EM–BP is more accurate in every one of the 192 scored cells — 16 seeds $\times$ 4 sizes $\times$ 3 regions $(all, bulk, centre)$, of which the table shows two. That count is descriptive: regions and sizes share fitted models and data, so the inferential unit behind each row is the 16 paired seeds. The ratio increases across the four sizes tested. That is a description of four measurements and not evidence that the residual is non-architectural: a baseline with an irreducible approximation floor b^2 gives $(b^2 + c_w/M)/(c_e/M)$, linear increasing in M for a wholly architectural reason, and the raw risks above are consistent with it — EM–BP's risk falls about fourfold across the range while the window head's falls by less than half that. Both arms select their budget symmetrically on a validation bundle disjoint from test, *within the grids on offer*: 22% of EM–BP cells and 27% of window-head cells take the last rung, which indicates validation error still falling where the grid ended rather than convergence. The values therefore compare the two procedures at stated, validation-selected budgets, not at converged optima. Appendix P reports the ratio under four aggregations of the same cells; all exceed $2\times$ at every size.

O Density-level recovery: the Hellinger figures

Both kernels are put on the same grid and renormalised under the same quadrature, so the comparison measures fit rather than normalisation convention. Absent a law over parent states a conditional kernel $K(\cdot \mid a)$ has no canonical scalar distance, and an unweighted median over grid columns gives a rarely-visited tail parent the same standing as a typical one — so it was, until this revision, a diagnostic on the fitted density rather than an intrinsic distance between transition laws. The intrinsic version is $\mathbb{E}_{A\sim\pi_0}[H(K_{\hat{\theta}}(\cdot \mid A), K_0(\cdot \mid A))]$ under the true parent law π_0, and both are reported below, computed by the same run (`outputs/exp_07_certified_clean_seed*/`, sixteen seeds, clean-deployed and gated; independently verified bit-identical to the certified table's own score-based fields, `outputs/README_exp07_certified_clean.md`).

M	32	64	128	256	512	1024	2048
relative score error	0.0580	0.0501	0.0342	0.0258	0.0220	0.0162	0.0132
transition Hellinger (unweighted median)	0.1030	0.0896	0.0757	0.0637	0.0598	0.0522	0.0500
transition Hellinger (parent-law-weighted mean)	0.0994	0.0867	0.0741	0.0630	0.0592	0.0521	0.0499

All three rows fall with M, so the transition is being recovered and not merely fitted. The weighted mean sits consistently a little below the unweighted median — the parent law downweights exactly the tail columns an unweighted vote overweights — without changing the qualitative story.

P Does the headline depend on how the ratio was averaged?

A ratio is a nonlinear function of two means, so its value depends on the order of averaging. Table 1 reports the mean of per-cell ratios, aggregated per seed. That is the right inferential unit — the twelve noise levels share a training set and a fitted model — but being the right unit does not make it the only defensible summary, so here is the same data under five.

Table 4: The headline under five estimands. (1) is what Table 1 reports. The interval is a paired bootstrap over seeds, 20,000 resamples; seeds are resampled rather than cells, because cells within a seed share a fitted model.

M	(1) per-cell	(2) ratio of means	(3) geometric	(4) median	(5) paired 95% CI
32	7.0	6.8	6.1	6.8	[6.1, 8.1]
64	7.8	6.6	6.7	6.6	[6.3, 9.5]
128	8.8	8.0	7.8	8.6	[7.9, 9.6]
256	9.7	8.8	8.6	9.2	[8.5, 10.9]
512	9.0	8.6	7.9	8.2	[8.1, 9.9]
1024	12.5	9.7	10.8	11.1	[10.5, 14.7]
2048	15.5	10.0	13.2	12.9	[13.0, 18.9]
4096	17.6	12.4	15.6	16.0	[15.4, 19.9]

Two things follow, and the second is the one worth stating plainly.

The conclusion does not depend on the choice: every estimand at every size exceeds 6.1, every bootstrap interval sits well clear of 1, and no aggregation produces a reversal anywhere.

But **the estimand we report is systematically the largest of the four**, and by a margin that grows with M — up to 5.5 at $M = 2048$. The ratio of per-seed averaged errors, which is the summary a reader is most likely to have in mind, runs 6.6 to 12.4 rather than 7.0 to 17.6.

That ordering is a measurement, not a theorem, and it is worth being precise about why no inequality delivers it. Jensen gives $\mathbb{E}[1/Y] \geq 1/\mathbb{E}[Y]$, but $\mathbb{E}[X/Y] \geq \mathbb{E}[X]/\mathbb{E}[Y]$ does *not* follow for arbitrary positive correlated X and Y — the dependence between numerator and denominator can push the comparison either way. What holds here is empirical: denominator heterogeneity across noise levels interacts with that dependence, and in this dataset the mean of cellwise ratios comes out largest. It is a property of these cells, which is exactly why the estimand has to be named rather than assumed.

Resolving the average over noise levels shows where the aggregation hides structure: the ratio is not flat in t but peaks near $t \approx 0.5$ and falls at both ends. Two different "weakest" quantities have to be kept apart. The weakest seed-averaged (M, t) aggregate in the grid is 3.6; the weakest *individual* cell is 0.92, and EM–BP is beaten outright in 2 of the $1,520$ cells summarised here. The estimator's advantage is largest at moderate noise, where the posterior is dominated by neither the likelihood nor the prior — which is where knowing the transition structure should help most.

The same question applies to the structure-aware ratio, and with more force, since that one is quoted as a range. There the noise schedule is already reduced into each cell's risk, so the seed is the inferential unit by construction and only the forming and pooling of the ratio remain optional.

Table 5: Table 3's all-site ratio under four aggregations of the same cells, with the paired bootstrap interval over seeds. (1) is what the table reports. The conclusion does not depend on the choice: every estimand at every size exceeds $1.98\times$, and the largest departure from the reported value is 0.48.

M	(1) per-seed	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (1)
32	2.34	2.00	2.15	1.98	[1.88, 2.86]
128	2.82	2.45	2.61	2.62	[2.29, 3.44]
512	3.58	3.43	3.51	3.59	[3.25, 3.91]
2048	6.14	5.66	5.89	5.93	[5.30, 7.02]